Response to Reviewer 2

We thank the reviewer for an exceptionally detailed and rigorous review, including the deep methodological critique. The review identified the most important weaknesses of the original submission: the overclaiming of state-of-the-art, the underpowered statistics, the unexplained Llama multi-step failure, the misleading $> 100\%$ metric, and the opacity of the generative model. The revision addresses each of these with new experiments and substantial new text. All locations below refer to the revised compiled PDF.

Summary of the main changes

- "State-of-the-art" has been removed; a direct EAP attribution ranking on the same backend is reported as the established-method comparison, and the absence of a portable ACDC run is documented as a limitation.
- Every reported number now carries a percentile bootstrap 95% confidence interval; H1 is not accepted where it is not significant under a paired permutation test (e.g. Llama IOI).
- The Llama multi-step failure is analysed and a finer-layer-bin remedy is attempted and reported (it does not close the gap), then stated as a major limitation.
- The $> 100\%$ metric is renamed Relative Cumulative KL (RCK); the ablation-only POMDP (POMDP-abl) against the ablation oracle is the primary discovery-quality comparison.
- The full A, B, C, D matrices and the exact Dirichlet concentrations are inlined; discretisation thresholds carry a $\pm 20\%$ sensitivity analysis; the 0.01 scaling factor is explained.
- An x86_64 (amd64) Docker image and a T4 run are provided.
- POMDP-abl is compared against the bandit (and a new plain UCB baseline) across all tasks, and a random-action baseline isolates EFE-guided action selection.

Major weaknesses and required revisions

> ***W1. Insufficient baseline comparisons and overclaiming of "state-of-the-art".*** *Run ACDC or EAP on a subset with a reasonable budget and report cumulative KL or faithfulness; otherwise state the limitation and remove "state-of-the-art".*

Response: The term "state-of-the-art" has been removed from the abstract and conclusion, replaced by "outperforms heuristic baselines". A direct Edge Attribution Patching ranking is now computed on the same `circuit-tracer` backend and reported on the common cumulative-KL / bounded-efficiency metric for both models (Table 2, p. 19; Figure 4, p. 20); EAP is the strongest baseline on both models. A genuine ACDC run is not included because ACDC is defined over the attention-head/MLP edge graph and does not operate on the transcoder-feature candidate set or the fixed per-feature budget used here; this is stated explicitly as a limitation of the present comparison (Discussion "Comparison scope and the absence of a direct ACDC run", p. 31).

> ***W2. Statistical weakness: small prompt sets and insignificant results.*** *Increase prompt sets to $n \geq 30$ or use bootstrapped confidence intervals; for Llama IOI accept that H1 is not supported. Investigate the Llama multi-step failure with per-prompt analysis and a remedy.*

Response: Every reported mean, bounded efficiency, and RCK now carries a percentile bootstrap 95% CI (10^4 resamples), and H1 is assessed with both a paired t-test and an exact paired-permutation test (Section "Statistical Analysis and Sensitivity", p. 17). H1 is explicitly not accepted for Llama IOI (Table 10, p. 30). The Llama multi-step failure is analysed per task with action-selection breakdowns; a finer-layer-bin variant was run as a remedy and reported even though it does not close the gap, and the failure is stated as a major limitation (Table 5, p. 26; Discussion, p. 31).

> ***W3. The "oracle efficiency*** $> 100\%$***" artefact is misleading.*** *Rename to "relative cumulative KL" and use the ablation-only POMDP as the primary comparison against the ablation oracle.*

Response: Done. Bounded oracle efficiency ($\leq 100\%$) is now defined only for ablation-only selectors and is the primary discovery-quality metric; the steering-enabled comparison is reported separately as RCK and labelled as a relative amplification factor (Section "Metrics", p. 16–17). POMDP-abl against the ablation oracle is the headline comparison throughout the IOI, multi-step, and domain tables.

> ***W4. Methodological opacity: generative model details missing.*** *Provide the full A, B, C, D matrices and Dirichlet concentrations; justify discretisation thresholds with* $\pm 20\%$ *sensitivity analysis; explain the* 0.01 *scaling factor.*

Response: The appendix "Complete Generative Model Specification" (p. 37) now inlines the full observation model A (all modalities), every action-conditioned B factor, the preference vector C, the prior D, and the exact initial Dirichlet concentrations ($pA = 10A$, learning rate $\eta = 1.0$). The $\pm 20\%$ threshold sweep is reported in Table 7 (p. 28), and the 0.01 scaling factor that maps graph importance into the KL-prior range is explained in the appendix and in Section "Statistical Analysis and Sensitivity" (p. 17).

> ***W5. Reproducibility and hardware constraints.*** *Provide a* `requirements.txt` *for x86_64 with a consumer GPU; report variance across hardware (T4 vs. DGX Spark); the Docker container targets aarch64.*

Response: An x86_64 (amd64) Docker image and a pinned `requirements.txt` are provided, and a T4 run is reported alongside the DGX Spark results to document hardware variance (Experimental Setup, p. 13 and reproducibility paragraph, p. 14). The environment and seeds are specified in the configuration.

> ***W6. Missing ablation for the action space and belief update.*** *Compare POMDP-abl vs. bandit across all tasks; add a random-action ablation that replaces the EFE objective to isolate its benefit.*

Response: A head-to-head table (Table 9, p. 29) now compares POMDP-abl against the bandit, the new plain UCB baseline, and EAP across IOI, multi-step, and the five domains for both models. A random-action baseline (EFE-free action selection with the same feature sampling) is reported on the RCK scale (Table 3, p. 22), isolating the contribution of EFE-guided action selection.

Minor issues

> *Terminology: "Oracle efficiency > 100%" should be renamed throughout.*

Response: Renamed to Relative Cumulative KL throughout, with bounded oracle efficiency reserved for the $\leq 100\%$ ablation-only quantity.

> *Figure 4: the text says "dominates all baselines", but on Llama the bandit tracks the POMDP closely; clarify.*

Response: The "dominates" claim has been removed. The text now states that EAP is the strongest baseline and that on Llama the bandit and POMDP-abl are close, consistent with Figure 4 (p. 20) and Table 2 (p. 19).

> *Table 2: high coefficient of variation; report median and quartiles.*

Response: Table 2 (p. 19) now reports the per-prompt median and inter-quartile range [Q1, Q3] in addition to the mean, for every method.

> *Section 3.3 (Actions): explain how the transition probabilities (50/25/25 etc.) were chosen.*

Response: The appendix (p. 37) now states that these encode how strongly each action is expected to move the latent importance state (ablation broad and near-symmetric, patching conservative, steering concentrated), and the $\pm 20\%$ threshold and hyperparameter sweeps (Tables 7 and 8, p. 28–29) show the conclusions are robust to the exact values.

> *Language: some sentences are overly long; a light copy-edit would improve readability.*

Response: The manuscript has been copy-edited for concision and a consistent third-person register throughout.

Deep methodology evaluation

> ***M1. Discretisation and state-space design.*** *The importance and layer-role binning is unjustified and architecture-dependent; the Llama multi-step failure is attributed to coarse discretisation; no sensitivity analysis; token position is ignored.*

Response: The discretisation is now justified and stress-tested. The $\pm 20\%$ threshold sweep (Table 7, p. 28) identifies the KL discretisation as the sensitive knob while leaving the qualitative ordering intact, and a finer-layer-bin variant is run for the Llama multi-step case (Table 5, p. 26). The absence of token-position factors is acknowledged as a limitation with a position-resolved observation model proposed as future work (Discussion "Token-position granularity", p. 31).

> ***M1 (cont.) Observation discretisation and online learning of*** A***.*** *Fixed KL/activation/connectivity thresholds are arbitrary; initial Dirichlet parameters are unspecified;* $\eta = 1.0$ *is unjustified; no belief-convergence evidence.*

Response: The exact thresholds, initial Dirichlet concentrations ($pA = 10A$), and $\eta = 1.0$ are now specified in the appendix (p. 37) and in the hyperparameter table. Online convergence of A is demonstrated with a new figure (Figure 9, p. 25) showing the mean per-step L_1 drift $\|A_t - A_{t-1}\|_1$ decreasing monotonically over the budget (roughly halving on Gemma IOI), and a hyperparameter sweep over η (Table 8, p. 29) shows the result is insensitive to the learning rate.

> ***M2. Generative model components** A, B, C, D. Full B matrices not shown; non-diagonal transitions underived; C assumes large effects are always better; D probabilities not given.*

Response: The full A, B, C, and D are inlined in the appendix (p. 37), including all non-diagonal B entries and the exact D probabilities. The risk that the preference model C rewards off-target or model-breaking interventions is now acknowledged directly (Discussion "Limitations", p. 31), and the steering analysis with a random-feature control and top-k token check (Results, p. 24) is the empirical response to that concern.

> ***M3. Candidate extraction and prior observation.** The top-5-per-layer selection biases the candidate set; the 0.01 prior-observation scaling is unexplained.*

Response: The candidate-selection procedure (top features by graph importance, up to five per layer) is now described precisely, and the resulting selection bias is acknowledged: the evaluation measures selection efficiency within an already-pruned set rather than over all model components (Discussion "Selection bias in the candidate set", p. 31). The 0.01 scaling, which maps graph importance into the KL-prior range, is explained in the appendix and Section "Statistical Analysis and Sensitivity" (p. 17).

> ***M4. Intervention selection via EFE.** Horizon-1 planning may miss synergies; sensitivity to $\omega_e = 2.0$ and $\gamma = 16$ is unexplored.*

Response: Single-step planning is retained but its limitation is stated explicitly, with tree-search planning proposed as future work (Discussion, p. 31). The action precision γ is now included in the hyperparameter sweep ($\gamma \in \{4, 16, 64\}$; Table 8, p. 29), which leaves bounded efficiency essentially unchanged. We clarify that $\omega_e = 2.0$ is the exploration term of the heuristic *bandit* baseline, not a POMDP parameter; the POMDP's exploration is the epistemic (information-gain) term of the EFE, and the new plain UCB baseline isolates the effect of the bandit's engineered priors.

> ***M5. Intervention engine and causal correctness.** The 10× multiplier is aggressive and may reflect model fragility, not causal role; no evidence it amplifies the intended concept; intervention information-leak across positions is not validated.*

Response: The 10× claim is now supported by a dose–response sweep ($m \in \{0, 1.5, 2, 3, 5, 10\}$) with a matched random-feature control, and by a top-k token analysis showing that large multipliers amplify coherent concept tokens rather than producing undifferentiated noise (Results "Feature Steering", p. 24). The causal claim is correspondingly softened to controllability. The intervention API operates on transcoder features at the final token position; the limitation that effects are not separately localised to earlier token positions is documented (Discussion "Token-position granularity", p. 31).

> ***M6. Baselines and comparative methodology.*** *ACDC/EAP comparison missing; the bandit's per-layer prior and one-visit decay are unjustified inductive bias; no simpler UCB baseline; POMDP-abl vs. bandit nuances undiscussed; random-action baseline missing.*

Response: A direct EAP ranking is reported (Table 2, p. 19; Figure 4, p. 20) and a plain UCB baseline was added that removes the bandit's engineered importance and locality priors, isolating how much of the bandit's performance comes from those priors versus upper-confidence exploration (Section "Baselines", p. 15; UCB rows in Table 2, p. 19). The POMDP-abl-versus-bandit nuances are now discussed across all tasks via the head-to-head table (Table 9, p. 29), and a random-action baseline isolates EFE-guided action selection (Table 3, p. 22). The portability limitation for ACDC is documented (Discussion, p. 31).

> ***M7. Reproducibility and implementation gaps.*** *Hardware dependency (aarch64); missing hyperparameters (number of runs, seeds, Dirichlet initialisation, exact B); code contents not described.*

Response: An x86_64 Docker image and a T4 run are provided, and the number of trials, all random seeds, the Dirichlet initialisation, η, γ, and the full B matrices are now specified (Experimental Setup, p. 13–14; appendix, p. 37). The experiment runner exposes every swept parameter through CLI flags, and the reported prompts correspond exactly to the released JSON results.

> ***M8. Recommendations for methodology revision.*** *Re-evaluate the layer-role discretisation; justify thresholds with empirical quantiles and ±20% sensitivity; provide full matrices; justify the steering multiplier; add ACDC/EAP and a random-action baseline; improve reproducibility (amd64 Docker, T4, seeds).*

Response: These recommendations are implemented as described in W1–W6 and M1–M7 above: the finer-layer-bin variant (Table 5), the ±20% and hyperparameter sweeps (Tables 7–8), the full inlined generative model (appendix), the dose–response steering justification (Results), the EAP comparison and random-action / UCB baselines (Tables 2, 3, 9), and the amd64 Docker image with a T4 run and specified seeds.

Final remark

The revision treats the Llama multi-step shortfall and the absence of a portable ACDC run as honest limitations rather than as resolved issues, and it confines all discovery-quality claims to the bounded ablation-only comparison so that no metric can exceed the oracle. We believe these changes address the methodological concerns raised.